CAP: Contrastive Activation Patching for Locating, Comparing, and Steering Neurons in Multilingual LLMs

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Abstract

Large language models routinely produce fluent output that is factually wrong or unsupported by the provided source, yet the neurons responsible for this behaviour remain poorly understood outside English. We present **CAP** (Contrastive Activation Patching), an open-source toolkit that *locates*, *compares*, and *steers* the neurons a model uses for a target behaviour across languages. From two contrastive groups of inputs, one command-line invocation captures per-neuron activations, ranks them with a Welch t-test and Benjamini–Hochberg FDR correction, and patches them to measure their causal effect on a benchmark; the same signal doubles as a training-free steering rule, and interactive viewers support per-layer inspection and cross-language comparison.

1 Introduction

Large language models routinely hallucinate, i.e., produce fluent output that is factually wrong or unsupported by the provided source (Ravichander et al., 2025). Hallucinations tend to be worse for medium- and low-resource languages, where data is sparser and evaluation less reliable (Ul Islam et al., 2025). *Mechanistic interpretability* (Ferrando et al., 2024) addresses such behaviour by locating the model components responsible for it, and one of its standard tools is *activation patching*, which intervenes on those components to test whether they are causally implicated (Meng et al., 2022). Applied to hallucination, however, this line of work is overwhelmingly English-centric, and the tooling to run such analyses *across languages* is largely absent; what exists tends to be single-experiment scripts rather than reusable tools.

CAP closes that gap. It packages *contrastive activation patching*, which compares a model's per-neuron activations on inputs it gets right versus wrong, as a reproducible, phase-resumable pipeline. The ingredients are established: activation patching as a causal test (Meng et al., 2022; Zhang and Nanda, 2024), and contrastive input pairs as a source of steering signal (Rimsky et al., 2024); CAP's contribution is to make them routine to run and compare across languages. Given two contrastive groups (*faithful* versus *unfaithful* text, correct versus incorrect predictions, or one task in two languages), it identifies the neurons that most distinguish them, tests whether they *causally* matter by patching, and reuses the same signal as a training-free steering intervention.

As a system, CAP makes three contributions. First, an end-to-end `cap` command-line interface that chains the four stages of an analysis (*capture* → *stats* → *patch* → *visualize*) and writes reproducible artefacts at each step: HDF5 activation tensors, a JSON run manifest, and phase-resumable stages, so that a long run can be interrupted and continued rather than restarted. Second, interactive HTML viewers that make the results inspectable without further code, covering both the per-layer distribution of a single run and the overlap between two runs, so that circuits can be compared across languages, tasks, and models. Third, a tested and documented Python package with bundled example data, so that the case study below can be reproduced and the pipeline applied to new behaviours and languages.

2 System and Method

The unit of analysis is a pair of contrastive prompt groups, supplied as one labelled CSV or two. A built-in template (FACTUALITY/FAITHFULNESS/NEGATION/raw) wraps each input so that its answer tokens expose the model's decision. From there, CAP runs four stages, each a `cap` sub-command:

- **Capture** runs one forward pass per group and streams per-neuron activations to HDF5, by default from the MLP down-projections: feed-forward layers act as key–value memories

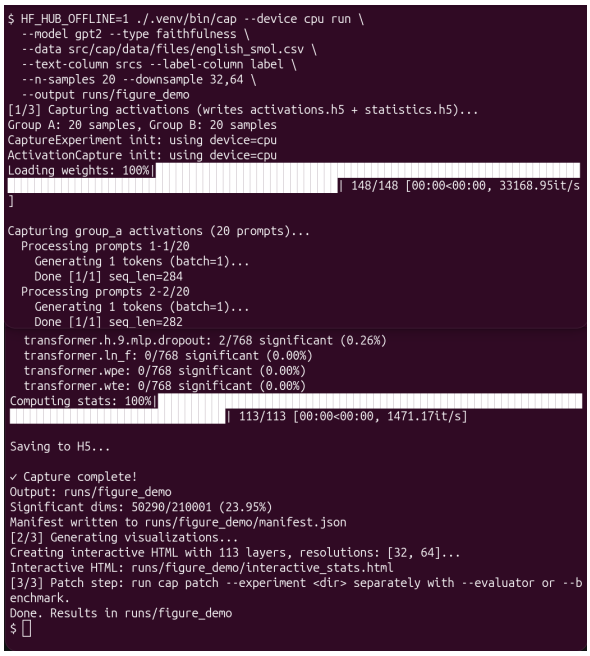


Figure 1: One command runs capture, stats and visualize, reporting the significant neurons and writing an interactive viewer.

whose neurons encode interpretable input patterns (Geva et al., 2021), which makes them a natural place to look for behaviour-specific computation.

- **Stats** asks which neurons separate the two groups, scoring each with a Welch t-test (Welch, 1947) and Cohen's d (Cohen, 1988) and correcting for multiple comparisons with Benjamini–Hochberg FDR (Benjamini and Hochberg, 1995). The neurons that survive form the *significant* set.
- **Patch** asks whether those neurons causally matter: it rescales them by a factor α (α=0 ablates) and re-evaluates the model on a benchmark, reporting the change in accuracy. Evaluators for GSM8K, MMLU, HellaSwag, and Multilingual MMLU (OpenAI, 2024) ship built in.
- **Visualize** renders a self-contained HTML: a *stats* viewer for the per-layer distribution of a single run (see Fig. 2), and a *similarity* viewer that overlays two runs and reports their per-layer agreement (for example, Jaccard overlap), so that comparing circuits across languages, tasks, or models reduces to loading the runs.

One `cap run` chains capture, stats, and visualize (see Fig. 1); `cap patch` adds the causal test.

3 Case Study: Qwen3 + Five Languages

We run CAP on Qwen3-8B and Qwen3-14B (Qwen Team, 2025) over SMOLDOC (Caswell et al., 2025; document-level factuality) and WMT MIST (Kocmi et al., 2025; cross-lingual summarisation), across English (EN), Egyptian Arabic (ARZ), Mandarin Chinese (ZH), German, and Spanish.

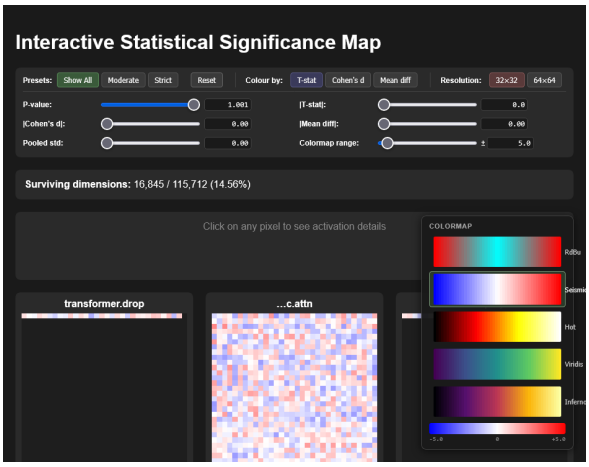


Figure 2: HTML interactive viewer.

We find that, *first*, factuality circuits are language-specific: of Qwen3-8B's 147,456 MLP down-projection neurons, 50.4% reach significance on SMOLDOC for EN but only 7.5% on ZH and 3.4% on ARZ (BH-FDR, α=0.05), a gap that narrows but does not vanish with scale.

Second, factuality and faithfulness are mechanistically distinct: the within-language overlap between SMOLDOC factuality and WMT MIST faithfulness neurons is indistinguishable from chance (Jaccard 0.0003 ARZ, 0.0017 ZH). This is the mechanistic counterpart to the human-evaluation case for splitting *hallucination* into *factuality* and *faithfulness* (Maynez et al., 2020; Schmidtova et al., 2025).

Third, steering transfers only when the source language is well chosen: as a training-free multiplicative intervention, ZH-sourced neurons on Qwen3-8B (α=1.04) improve all three SMOLDOC targets (+2.7, +3.1, +1.6 pp on EN, ARZ, ZH) at near-zero Multilingual MMLU cost, while on 14B the best source shifts to ARZ and a poorly chosen one is cross-lingually destructive.

4 Availability

CAP is MIT-licensed with bundled data, examples, tests, and docs. It requires Python $\geq$ 3.10 and PyTorch 2.x. Code and screencast: `https://github.com/Humeruzz/cap`, `https://streamable.com/ywsjtf`.

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